Cornell AAP
City and Regional Planning

To: Panamá City Resilience Office & 100 Resilient Cities
From: Tomás Schmieder
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Subject: Strengthening Panamá City's Climate Adaptation

Executive Summary

Panamá City's *Resilience Strategy* (2019) outlines an inclusive vision but lacks the data systems and monitoring standards needed to turn that vision into measurable climate adaptation. The plan itself identifies the need for shared environmental data and standards to measure resilience progress and guide investments. This absence prevents coordination, evaluation, and access to long-term resilience finance. Drawing on Barnett & O'Neill (2010), adaptation efforts that proceed without data risk maladaptation, actions that unintentionally heighten exposure to inequality. Ayers & Dodman (2010) emphasizes that successful adaptation depends on institutional learning and integration across development sectors. To achieve this, it is recommended that Panamá City work to establish a collaborative platform, linking municipal agencies, universities, and communities to transform the strategy into a data-driven, equity-focused governance tool.

Background

Situated on a low-lying tropical coast, Panamá City faces intensifying flood, storm, and heat risks. Urban expansion into reclaimed coastal zones, combined with deteriorating drainage systems, has increased exposure of both formal and informal settlements. While the *Resilience Strategy's* five pillars—opportunity access, equitable neighborhoods, coexistence with water, risk management, and shared city-building—capture essential priorities, they remain qualitative. No spatial risk baselines, climate projections, or performance indicators accompany them. The plan itself acknowledges fragmented environmental information and weak coordination among agencies such as MiAmbiente (Panamá's Ministry of Environment), SENACYT (National Secretariat for Science, Technology, and Innovation), and SINAPROC (Civil Protection). The document admits that "Local authorities do not have a full map of risks in the urban basins; it requires a comprehensive, validated tool to integrate the map of risks to the ordering of plans and thus foster an informed decision-making process" (p. 118), directly highlighting the gap between its vision and current capabilities.

Critical Assessment

Panamá City's current framework contrasts sharply with what should be happening: a data-driven system that link scientific observation, policy design, and community input through continuous feedback. Ayers & Dodman (2010) describe this model as "adaptive governance," an iterative process that evolves through learning and cross-sectoral collaboration. This would enable decision-makers to identify trade-offs and adjust priorities as conditions change.

The cause for this gap appears rooted in the 100 Resilient Cities model, which privileges visionary narratives over operational tools for empirical monitoring and implementation. This top-down approach can be problematic, Anguelovski et al. (2020) warns that technocratic "green" planning

often reinforces inequities when social realities are excluded from data production. As a result, Panamá City's strategy, while inclusively mentioning high risk communities, lacks mechanisms for communities to generate or verify environmental data, leaving marginalized communities possibly invisible in decision-making.

The effect of this data and participation gap is a heightened risk of maladaptation. As Barnett & O'Neill (2010) caution, adaptation without evidence can develop into investments that worsen vulnerability or constrain future options. If the city continues implementing projects without an integrated information base, it risks funding initiatives that fail to aid vulnerable communities or anticipate the accelerating impacts of climate change.

Recommendations

To translate the *Resilience Strategy* into measurable actions, Panamá City should prioritize coordination, community participation, and enforceable regulation. These steps address the current data and governance gaps that limit adaptive learning and equity in decision-making. The recommended actions are as follows:

1. Establish a climate-risk data platform managed by the Municipality, in partnership with MiAmbiente, SENACYT, and local universities. The platform would centralize and standardize hydrological, meteorological, and socioeconomic data, producing indicators that enable continuous assessment and policy evaluation.
2. Co-produce vulnerability maps, particularly with residents of flood-prone basins or lower income communities. Municipal and academic partners could train local organizations to document infrastructure failures and hazards, ensuring that local knowledge informs adaptation priorities.
3. Link data to enforceable spatial policy by using monitoring outputs to update zoning, drainage standards, and building codes. Pairing new regulations with data helps prevent maladaptation and ensures that future planning anticipates evolving climate risks.

Together, these measures would embed adaptative governance within Panamá City's institutions, connecting technical data with lived experience, allowing for a transformation of the strategy from an aspirational goal to a transparent, evidence-based practice that attracts sustained climate investment.

Conclusion

Panamá City's climate future depends on the assimilation of new systems that learn, adapt, and earn public trust. Establishing an integrated data platform, participatory mapping, and enforceable spatial policy can shift the city from reactive disaster management to proactive adaptation. However, such transitions must be implemented carefully. Stronger regulation and data transparency could lead to problems such as raising property costs or deepening distrust of communities excluded from decision-making. To avoid such outcomes, implementation must be gradual, transparent, and grounded in partnership with affected residents. As Ayers & Dodman (2010) emphasize, adaptation succeeds through learning and integration across sectors, while Barnett & O'Neill (2010) caution that poorly designed interventions can lead to maladaptation. If pursued inclusively, these measures can transform the *Resilience Strategy* from vision to measurable progress.

References

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